1. EQUIPMENT IDENTITY AND PURPOSE

Machine ID: M-AC90
Model: AC-90
Description: Industrial rotary screw air compressor.
The AC-90 is an industrial air compressor designed to provide steady pressurized plant air to production tooling.
Lubricant specification: AC-L32 compressor lubricant (Note: Never use H46 hydraulic oil in this equipment).

2. OPERATING REFERENCE VALUES

- Normal outlet pressure: 7.0 to 8.0 bar under steady demand in LOADED state.
- Low-pressure warning threshold: Below 6.5 bar for more than 10 seconds.
- Normal discharge temperature: 70°C to 90°C.
- High-temperature warning threshold: Above 95°C.
- High-temperature stop threshold: 105°C or higher (thermal trip).
- Lubricant level: Must remain between marked LOW and HIGH level sight indicators.

3. MACHINE OPERATING STATES

- READY: Preconditions satisfied, standby mode awaiting air demand.
- LOADED: Compressor actively delivering compressed air under system demand.
- HOLD: Operation interrupted or locked out for inspection.

Note: The normal outlet pressure of 7.0-8.0 bar applies specifically under steady demand in the LOADED state. A temporary transient pressure dip does not constitute an alarm condition.

1. ALARM CODES REFERENCE & DIAGNOSTIC LIMITS

- E17: High discharge temperature. On AC-90, E17 indicates discharge air temperature exceeding limits. (Normal: 70-90°C; Warning: Above 95°C; Stop: 105°C or higher). Required response: Record discharge temperature; place machine on HOLD if above 95°C; request cooling-air path review.

CRITICAL DISTINCTION: On AC-90, E17 means high discharge temperature. On HP-200, E17 means low hydraulic pressure. Never mix these machine models.

- E08: Low outlet pressure. Triggered when outlet pressure remains below 6.5 bar for more than 10 seconds under steady demand. Required response: Record pressure, duration, demand state, and check for audible/visible air leakage; inspect AF-10 inlet-filter restriction indicator. Note: E08 does not prove that the compressor airend element is worn.
- E31: Condensate-drain fault. Indicates automatic condensate-drain failure. (Contrast: On HP-200, E31 is guard-open). Required response: Record fault and request authorized drain inspection.
- E42: Inlet-filter restriction. Indicates restriction in the AF-10 compressor air inlet filter. Required action: Record indicator and review AF-10 service history.

2. TEMPERATURE RESPONSE EXAMPLES

If discharge temperature reaches 98°C:

- 98°C is above the 95°C warning threshold, but below the 105°C shutdown stop threshold.
- The compressor must remain on HOLD and the cooling air path must be inspected.

1. SCHEDULED MAINTENANCE INTERVALS
- Before each shift: Check AC-L32 lubricant level (between LOW and HIGH) and verify condensate-drain status.
- Every 100 operating hours: Inspect the AF-10 air inlet-filter restriction indicator. Record state.
- Every 400 operating hours: Inspect external cooling-air path for blockages and cleanliness.
- Every 600 operating hours: Planned replacement of AF-10 inlet filter. (Earlier replacement required if restriction confirmed).
- Every 3,000 operating hours or 12 months: Planned replacement of AC-L32 compressor lubricant (whichever occurs first).

2. INSPECTION VERSUS REPLACEMENT RULES
- For AF-10 inlet filter: Restriction indicator must be inspected every 100 operating hours. Planned replacement is scheduled every 600 operating hours.
- Note: The manual does not provide a calendar-based replacement interval for the AF-10 filter.

3. ESCALATION & RETURN-TO-SERVICE REQUIREMENTS
When an AC-90 abnormal condition occurs, record: Machine ID (M-AC90), timestamp, operating hours, active alarm, outlet pressure, discharge temperature, air demand state, lubricant sight-glass level, AF-10 filter indicator, and condensate-drain condition.
Physical maintenance requires authorized personnel using site pressure-release and isolation procedures (not contained in manual).
Return-to-service authorization: The designated supervisor alone records the return-to-service decision.

4. EXCLUDED INFORMATION
The manual does not provide: maintenance costs, spare parts pricing, supplier contacts, electrical schematics, bolt torque ratings, motor rewinding specifications, compressor element overhaul steps, expected remaining service life, or guaranteed repair times.